3

Univariate Population Parameters and Sample Statistics

Chapter Outline

3.1 Summation Notation
3.2 Measures of Central Tendency
  3.2.1 The Mode
  3.2.2 The Median
  3.2.3 The Mean
  3.2.4 Summary of Measures of Central Tendency
3.3 Measures of Dispersion
  3.3.1 The Range
  3.3.2 *H* Spread
  3.3.3 Deviational Measures
  3.3.4 Summary of Measures of Dispersion
  3.3.5 Recommendations Based on Measurement Scale
3.4 Computing Sample Statistics Using SPSS
  3.4.1 Explore
  3.4.2 Descriptives
  3.4.3 Frequencies
3.5 Computing Sample Statistics Using R
  3.5.1 Reading Data into R
  3.5.2 Generating Sample Statistics
3.6 Research Question Template and Example Write-Up
3.7 Additional Resources

Key Concepts

1. Summation
2. Central tendency
3. Outliers
4. Dispersion
5. Exclusive versus inclusive range
6. Deviation scores
7. Bias

In Chapter 2, we began our discussion of descriptive statistics, previously defined as techniques that allow us to tabulate, summarize, and depict a collection of data in an abbreviated fashion. We considered various methods for representing data for purposes of communicating something to the reader or audience. In particular, we were concerned with ways of representing data in an abbreviated fashion through both tables and figures.

In this chapter, we delve more into the field of descriptive statistics in terms of three general topics. First, we examine **summation notation**, which is important for much of the chapter, and to some extent, the remainder of the text. Second, *measures of central tendency* allow us to boil down a set of scores into a single value, a point estimate, which somehow represents the entire set. The most commonly used measures of central tendency are the mode, median, and mean. Finally, *measures of dispersion* provide us with information about the extent to which the set of scores varies—in other words, whether the scores are spread out quite a bit or are pretty much the same. The most commonly used measures of dispersion are the range (exclusive and inclusive ranges), *H* spread, and variance and standard deviation. In summary, concepts to be discussed in this chapter include summation, central tendency, and dispersion. Within this discussion, we also address outliers and bias. Our objectives are that by the end of this chapter, you will be able to do the following: (a) understand and utilize summation notation, (b) determine and interpret the three commonly used measures of central tendency, and (c) determine and interpret commonly used measures of dispersion.

3.1 Summation Notation

A superbly talented and motivated group of graduate students are working in the statistics lab. We now find Oso Wyse tasked with his first lead role.

The graduate students in the statistics lab, Addie Venture, Oso Wyse, Challie Lenge, and Ott Lier, have been assigned their first task as research assistants. Dr. Debhard, a statistics professor, has given the group of students quiz data collected from 25 students enrolled in an introductory statistics course and has asked the group to summarize the data. Dr. Debhard was pleased with the descriptive analysis and presentation of results previously shared and is now working with Oso Wyse to conduct additional analyses related to the following research questions: *How can quiz scores of students enrolled in an introductory statistics class be summarized using measures of central tendency? How can quiz scores of students enrolled in an introductory statistics class be summarized using measures of dispersion?*

Many areas of statistics, including many methods of descriptive and inferential statistics, require the use of summation notation. Say we have collected heart rate scores from 100 students. Many statistics require us to develop "sums" or "totals" in different ways. For example, what is the simple sum, or total, of all 100 heart rate scores? **Summation** (i.e., addition) is not only quite tedious to do computationally by hand, but we also need a system of notation to communicate how we have conducted this summation process. This section describes such a notational system.

3.2 Measures of Central Tendency

One method for summarizing a set of scores is to construct a single index or value that can somehow be used to represent the entire collection of scores. In this section we consider the three most popular indices, known as **measures of central tendency**. Although other indices exist, the most popular ones are the mode, the median, and the mean.

3.2.1 The Mode

The simplest method to use for measuring central tendency is the mode. The **mode** is defined as *that value in a distribution of scores that occurs most frequently*. An easy way to remember the definition of the mode is to associate "mode" with "most," as that what the mode represents—the value or category (in the case of nominal or ordinal variables) that occurs most often. Consider the example frequency distributions of the number of hours of TV watched per week, as shown in Table 3.1. In distribution *f*(a) the mode is easy to determine, as the interval for value 8 contains the most scores, three (i.e., the mode number of hours of TV watched is 8). In distribution *f*(b) the mode is a bit more complicated as two adjacent intervals each contain the most scores; that is, the 8- and 9-hour intervals each contain three scores. Strictly speaking, this distribution is *bimodal*, that is, containing two modes, one at 8 and one at 9. This is our personal preference for reporting this particular situation. However, because the two modes are in adjacent intervals, some individuals make an arbitrary decision to average these intervals and report the mode as 8.5.

Distribution *f*(c) is also bimodal; however, here the two modes at 7 and 11 hours are not in adjacent intervals. Thus, one cannot justify taking the average of these intervals, as the average of 9 hours, 7+11)/2, is not representative of the most frequently occurring score. The score of 9 occurs less than any other score observed. We recommend reporting both modes here as well. Obviously, there are other possible situations for the mode (e.g., trimodal distribution), but these examples cover the basics. As one further example, the example data on the statistics quiz from Chapter 2 are shown in Table 3.2 and are used to illustrate the methods in this chapter. The mode is equal to 17 because that interval contains more scores (five) than any other interval. Note also that the mode is determined in

TABLE 3.1

Example Frequency Distributions

X	*f*(a)	*f*(b)	*f*(c)
6	1	1	2
7	2	2	3
8	3	3	2
9	2	3	1
10	1	2	2
11	0	1	3
12	0	0	2

Measure of Central Tendency	Advantages	Disadvantages
Mean	Function of all scores in the distribution	Influenced by extreme scores
	Has a unique value	Cannot be used with nominal or ordinal variables
	Easy to deal with mathematically	
	Can be used with interval and ratio measurement scales of variables	

We began our discussion of measures of central tendency by stating that *other indices exist*; however, the most popular ones are the mode, the median, and the mean. You may be wondering what those other indices are! While the *arithmetic mean* is the most common mean, and the one with which we are generally concerned, it is not the only mean, and it should not be confused with other types of means. Other means that you may encounter include the harmonic mean, trimmed mean, winsorized mean, and more. Huck (2016) provides a concise discussion on understanding the most common measures of central tendency relative to other statistics you may encounter.

3.3 Measures of Dispersion

In the previous section, we discussed one method for summarizing a collection of scores, the measures of central tendency. Central tendency measures are useful for describing a collection of scores in terms of a single index or value (with one exception: the mode for distributions that are not unimodal). However, what do they tell us about the distribution of scores? Consider the following example. If we know that a sample has a mean of 50, what do we know about the distribution of scores? Can we infer from the mean what the distribution looks like? Are most of the scores fairly close to the mean of 50, or are they spread out quite a bit? Perhaps most of the scores are within 2 points of the mean. Perhaps most are within 10 points of the mean. Perhaps most are within 50 points of the mean. Do we know? The answer, of course, is that the mean provides us with no information about what the distribution of scores looks like, and any of the possibilities mentioned, and many others, can occur. The same goes if we only know the mode or the median.

Another method for summarizing a set of scores is to construct an index or value that can be used to describe the amount of *spread* amongst the collection of scores. In other words, we need measures that can be used to determine whether the scores fall fairly close to the central tendency measure, are fairly well spread out, or are somewhere in between. In this section we consider the four most popular such indices, which are known as **measures of dispersion** (i.e., the extent to which the scores are dispersed or spread out). Although other indices exist, the most popular ones are the range (exclusive and inclusive), *H* spread, variance, and standard deviation.

3.3.1 The Range

The simplest measure of dispersion is the **range**. The term *range* is one that is in common use outside of statistical circles, so you have some familiarity with it already. For instance,

(continued)

Measure of Dispersion	Advantages	Disadvantages
H spread	Unaffected by extreme scores Can be used with ordinal, interval, and ratio measurement scales of variables	Not a function of all scores in the distribution Difficult to deal with mathematically due to its instability Cannot be used with nominal data
Variance and standard deviation	Function of all scores in the distribution Useful for deriving other statistics Can be used with interval and ratio measurement scales of variables	Influenced by extreme scores Cannot be used with nominal or ordinal variables

3.3.5 Recommendations Based on Measurement Scale

A summary of when these descriptive statistics are most appropriate for each of the scales of measurement is shown in Box 3.3. Throughout the text we emphasize that it is the researcher's responsibility to understand the data and its measurement scale so that the appropriate statistics can be generated given the measurement scale of the data.

BOX 3.3 Appropriate Descriptive Statistics

Measurement Scale	Measure of Central Tendency	Measure of Dispersion
Nominal	Mode	
Ordinal	Mode	Range
	Median	*H* spread
Interval/ratio	Mode	Range
	Median	*H* spread
	Mean	Variance and standard deviation

3.4 Computing Sample Statistics Using SPSS

The purpose of this section is to see what SPSS has to offer in terms of computing measures of central tendency and dispersion. In fact, SPSS provides us with many different ways to obtain such measures. The three tools that we have found to be most useful for generating descriptive statistics covered in this chapter are Explore, Descriptives, and Frequencies.

3.4.1 Explore

Step 1. The first tool, Explore, can be invoked by clicking "Analyze" in the top pulldown menu, then "Descriptive Statistics," and then "Explore." Following the screenshot in Figure 3.1 will produce the Explore dialog box. For brevity, we have not reproduced this initial screenshot when we discuss the Descriptives and Frequencies programs; however, you can see in Figure 3.1 where they can be found on the pulldown menus.

3.5 Computing Sample Statistics Using R

Next we consider **R** for computing the mean, median, mode, standard deviation, variance, range, minimum, and maximum. The commands are provided within the blocks with additional annotation to assist you in understanding how each command works. Should you want to write reminder notes and annotation to yourself as you write the commands in **R** (and we highly encourage doing so), remember that any text that follows a hashtag (i.e., #) is annotation only and not part of the **R** code. Thus, you can write annotations directly into **R** with hashtags. We encourage this practice so that when you call up the commands in the future you'll understand what the various lines of code are doing. You may think you'll remember what you did. However, trust us. There is a good chance that you won't. Thus, consider it best practice when using **R** to annotate heavily!

3.5.1 Reading Data into R

We will first read in our data (Figure 3.7). We will be using the quiz data that we have used previously.

```
getwd()
```

R is always pointed to a directory on your computer. To find out which directory it is pointed to, run this "get working directory" command. We will assume that we need to change the working directory, and will use the next line of code to point the working directory to the desired path.

```
setwd("E:/Folder")
```

To set the working directory, use the *setwd* function and change what is in quotation marks here to your file location. Also, if you are copying the directory name, it will copy in slashes. You will need to change the slash (i.e., \) to a forward slash (i.e., /). Note that you need your destination name within quotation marks in the parentheses.

```
Ch3_quiz <- read.csv("Ch3_quiz.csv")
```

The *read.csv* function reads your data into **R**. What's to the left of the <- will be what you want to call the data in **R**. In this example, we're calling the dataframe "Ch3_quiz." What's to the right of the <- tells **R** to find this particular .csv file. In this example, our file is called "Ch3_quiz.csv." Make sure the extension (i.e., .csv) is included in your script. Also note that you need the name of your file in quotation marks within the parentheses.

```
names(Ch3_quiz)
```

The *names* function will produce a list of variable names for each dataframe as follows. This is a good check to make sure your data have been read in correctly.

```
[1] "quiz"
```

```
view(Ch3_quiz)
```

The *View* function will let you view the dataset in spreadsheet format in RStudio.

FIGURE 3.7
Reading data into **R**.

```
     SE.mean CI.mean.0.95          var
  0.6326663    1.3057591   10.0066667

     std.dev     coef.var
  3.1633316    0.2032989
```

FIGURE 3.9 (continued)
Summary statistics using the *pastecs* package.

If we want to produce just one statistic, such as the mean, standard deviation, or variance, we can generate those values with the following scripts (Figure 3.10). The first part of the script defines the function (i.e., *mean, sd, var*) to compute the mean, standard deviation, or variance, respectively. What is enclosed in parentheses tells **R** which dataframe (i.e., "Ch3_quiz") and which variable within that dataframe (i.e., "quiz") to use to compute the statistics. These terms are separated by a $.

```
mean(Ch3_quiz$quiz)
sd(Ch3_quiz$quiz)
var(Ch3_quiz$quiz)
```

The *mean*, *sd*, and *var* functions can be used to generate, respectively, the mean, standard deviation, and variance. The script *Ch3_quiz$quiz* tells **R** to use the variable "quiz" from the "Ch3_quiz" dataframe. The output follows.

```
# mean(Ch3_quiz$quiz)
[1] 15.56

# sd(Ch3_quiz$quiz)
[1] 3.163332

# var(Ch3_quiz$quiz)
[1] 10.00667
```

FIGURE 3.10
Sample statistics.

3.6 Research Question Template and Example Write-Up

As we stated in Chapter 2, depending on the purpose of your research study, you may or may not write a research question that corresponds to your descriptive statistics. If the end result of your research paper is to present results from inferential statistics, it may be that your research questions correspond only to those inferential questions, and thus no question is presented to represent the descriptive statistics. That is quite common. On the other hand, if the ultimate purpose of your research study is purely descriptive in nature, then writing one or more research questions that correspond to the descriptive statistics is not only entirely appropriate but (in most cases) absolutely necessary. At this time, let us revisit our graduate research assistant, Oso Wyse, who was working with Dr. Debhard. As you may recall, his task was to summarize data from 25 students enrolled in a statistics course. The questions with which Oso was assisting Dr. Debhard were as follows: *How can quiz scores of students enrolled in an introductory statistics class be summarized using measures*

of central tendency? How can quiz scores of students enrolled in an introductory statistics class be summarized using measures of dispersion?

The following is a template for writing descriptive research questions for summarizing data with measures of central tendency and dispersion:

How can [variable] be summarized using measures of central tendency? How can [variable] be summarized using measures of dispersion?

Next, we present an APA-like paragraph summarizing the results of the statistics quiz data example answering the questions posed to Marie.

As shown in Table 3.5, scores ranged from 9 to 20. The mean was 15.56, the approximate median was 17.00 (or 16.33 when calculated from grouped data), and the mode was 17.00. Thus, the scores tended to lump together at the high end of the scale. A negatively skewed distribution is suggested given that the mean was less than the median and mode. The exclusive range was 11, *H* spread (interquartile range) was 5.0, variance was 10.007, and standard deviation was 3.1633. From this we can tell that the scores tended to be quite variable. For example, the middle 50% of the scores had a range of 5 (*H* spread), indicating that there was a reasonable spread of scores around the median. Thus, despite a high "average" score, there were some low-performing students as well. These results are consistent with those described in Section 2.4.

3.7 Additional Resources

In the previous chapters, we have mentioned a number of excellent resources for learning statistics. As we are still in the early stages of learning statistics, there are no additional resources that we suggest here. Rather, we refer you back to those chapters for supplemental resources for statistics as well as statistical software.

Problems

Conceptual Problems

1. Adding just one or two extreme scores to the low end of a large distribution of scores will have a greater effect on:

a. *Q* than on the variance.

b. the variance than on *Q*.

c. the mode than on the median.

d. none of the above.